

Free Water Access Under Urban Heat Stress: Mapping Drinking Fountain Accessibility and Climate Vulnerability in Italian Cities

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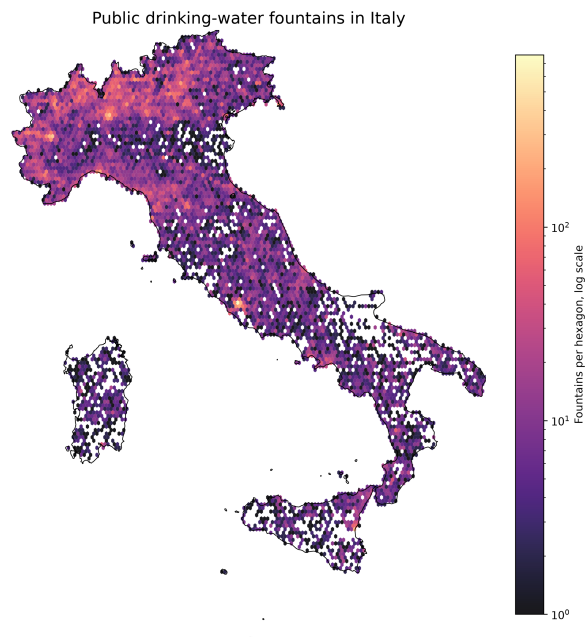
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ABSTRACT: This project evaluates whether Italian cities' free public drinking-water infrastructure is spatially aligned with present and future heat-health risk. Combining mapped fountains, walking-distance accessibility, heatwave indicators, urban heat exposure, demographic vulnerability, and available health-burden data, the project identifies neighborhoods where climate adaptation through free water access is insufficient.

Contents

1	Introduction	1
2	Fountain accessibility	1
3	Heat exposure and vulnerability	1
4	Data analysis	2
4.1	Turin	3
4.2	Milan	3
4.3	Rome	3
4.4	Naples	3

1 Introduction



2 Fountain accessibility

Share of population within 250m or 500m walking distance of a fountain.

- fountains per km^2
- fountains per 1000 residents
- fountains per 1000 elderly residents
- median walking distance to nearest fountain
- share of residents within 250 m or 500 m
- share of vulnerable residents within 250 m or 500 m
- correlation with average income distribution

3 Heat exposure and vulnerability

For each city or neighborhood, compute:

- historical heatwave days;
- tropical nights;
- annual maximum temperature;
- projected increase in maximum temperature;
- projected increase in heatwave days;
- elderly population share;
- socioeconomic deprivation;
- impervious surface;
- tree cover;
- urban heat island intensity.

4 Data analysis

Compute correlation matrices between variables in the previous sections. We expect there may be non-uniform mapping effects: we could repeat the analysis for urban areas such as

4.1 Turin

4.2 Milan

4.3 Rome

4.4 Naples

Acknowledgements

References